

Lusitano Fabric Defect Detection Dataset

[Home](#) [Dataset/Downloads](#) [Experiments](#) [About](#)

Home

Our efforts are aimed at addressing the intricate challenges posed by the extensive diversity in colors, textures, and defect characteristics encountered in fabric defect detection. To facilitate research and innovation in this field, we are pleased to introduce a groundbreaking dataset. This dataset comprises a comprehensive selection of fabrics and defects sourced from a reputable textile company based in Portugal.

Dataset/Downloads

The Lusitano dataset was collected over a 3-month period, spanning from January to March, from Paulo de Oliveira, S.A., a prominent textile company, based in Covilhã, Portugal, renowned for its innovative contributions to the textile industry. To collect the images for the dataset, we placed one camera in front of a fabric inspection machine, along with a strong and nearly uniform light source. This dataset comprises 4096×1024 images, captured by an industrial-grade Teledyne Dalsa Linea camera. The camera's high resolution and precision ensure the accurate depiction of textile samples, with the level of detail necessary for defect analysis. None of the defects depicted in this dataset are artificially generated; they stem from genuine occurrences

Lusitano Fabric Defect Detection Dataset

[Home](#) [Dataset/Downloads](#) [Experiments](#) [About](#)

architecture, a MVTEC AD dataset. The training set contains 32k normal images, while the testing set contains 1646 defect images and 1038 normal images.



Real-time setup image from factory showing camera and fabric

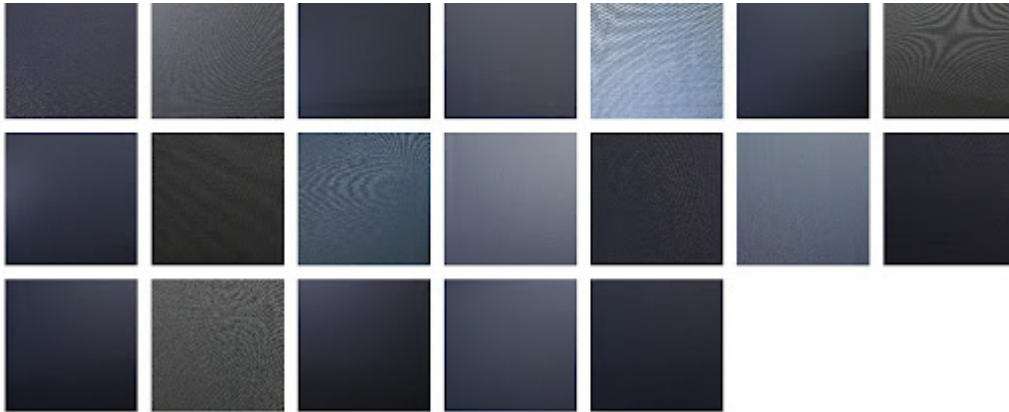
Lusitano Fabric Defect Detection Dataset

[Home](#)

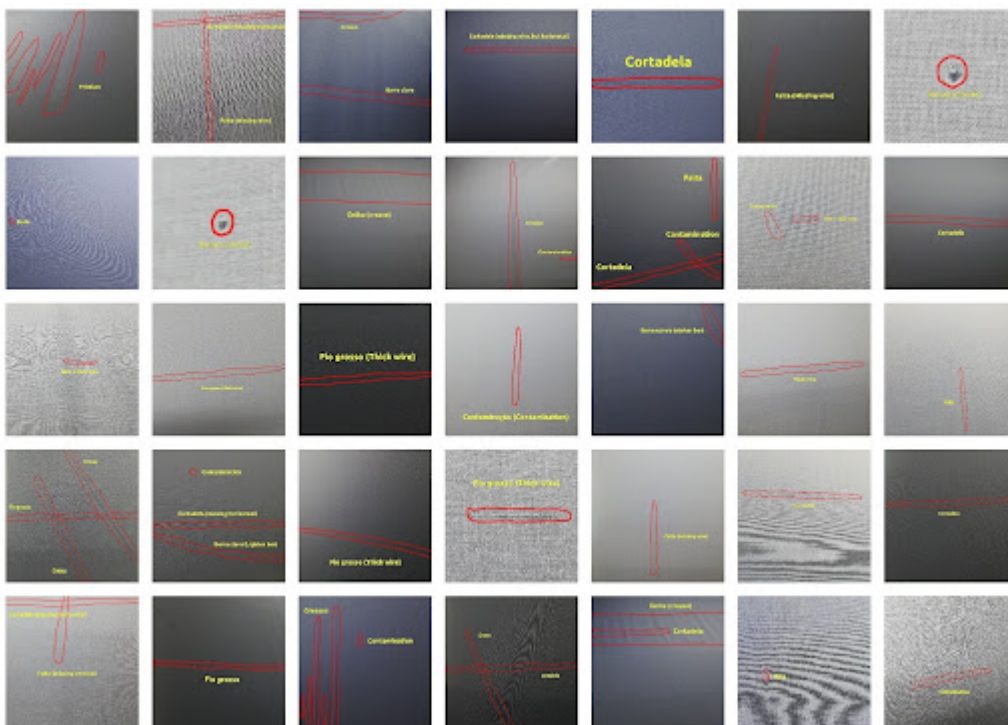
Dataset/Downloads

Experiments

About



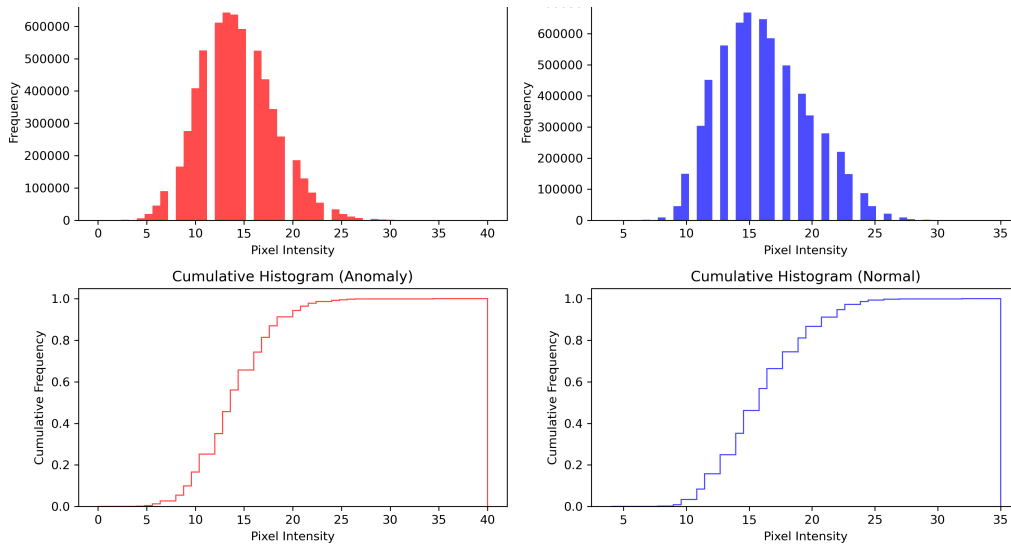
Normal Fabric Images: A collection of high-resolution images showing various examples of normal fabric textures. These images serve as reference samples for fabric quality assessment.



Anomaly Fabric Images: A collection of fabric defect examples collected by our line camera. Defect areas are marked for reference.

Lusitano Fabric Defect Detection Dataset

[Home](#) [Dataset/Downloads](#) [Experiments](#) [About](#)



Comparison of defect and normal images along with their pixel-wise and cumulative histograms.

Train Dataset.V2(6GB) (Note this is second v2) having 6000 non defects images for training)

Test Dataset (12.5GB)

Experiments

Method Used	Sample Size	AUC	Change (%)	mAP	Change (%)
DBFAD [91]	1K	0.7573	-	0.8162	-
	2K	0.7570	-0.04%	0.8199	0.45%
	4K	0.7691	1.56%	0.8293	1.60%
	8K	0.7737	2.17%	0.8301	1.70%
	16K	0.7797	2.96%	0.8354	2.35%
	32K	0.7814	3.18%	0.8429	3.27%
RD4AD [92]	1K	0.8231	-	0.8773	-
	2K	0.8189	-0.51%	0.8803	0.34%
	4K	0.8640	4.97%	0.9275	5.72%
	8K	0.8726	6.01%	0.9319	6.22%
	16K	0.8831	7.29%	0.9374	6.85%
	32K	0.8860	7.64%	0.9390	7.03%

Citations:

Lusitano Fabric Defect Detection Dataset

[Home](#) [Dataset/Downloads](#) [Experiments](#) [About](#)

About

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Lusitano Fabric Defect Detection Dataset

[Home](#) [Dataset/Downloads](#) [Experiments](#) [About](#)

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